14.12.2 Objectives

By the end of this lab, you will:

- Activate API support in Laravel 12
- Set up API routes using Laravel 12
- Use Laravel API Resource Controllers for CRUD operations
- Connect Laravel 12 to a MySQL database
- Implement request validation in Laravel
- Test the API using Postman or cURL

14.12.3 Prerequisites

Before starting, ensure you have:

- **PHP 8.2+**, Composer, and MySQL installed
- **Laravel 12 installed**
- **Postman or cURL** for API testing
- **A MySQL database already set up**

> *This lab uses **PHP 8.4**, **Laravel 12** and **Visual Studio Code** as the primary editor.*

Let’s get started!

14.12.4 Steps

Here are the steps to complete this lab:

14.12.4.1 Step 1: Create Laravel Project

Open your terminal and create a new Laravel project:

```
laravel new product-api
cd product-api
code .
```

You should see the Laravel project structure in your code editor.

14.12.4.2 Step 2: Configure the Database Connection

Open the `.env` file and configure the MySQL connection: